

Hasan Bukhari

Machine Learning • Research Engineering • Scientific Computing

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EDUCATION

University of Southern Mississippi

B.S. in Computer Science; Minor in Biological Sciences

GPA: 4.00 • President's List • Academic Excellence Scholar

Beaconhouse College Program

Cambridge A Levels: 4 As in Biology, Chemistry, Mathematics, and Physics

Expected Dec 2028

Hattiesburg, MS

May 2024

Karachi, Pakistan

MACHINE LEARNING & RESEARCH EXPERIENCE

Independent Computational Biology Researcher

2026 – Present

CancerShift and Chromatin Contact Prediction Across Chromosomes

Independent

- **CancerShift:** Developed a leakage-controlled multicancer classification pipeline spanning TCGA, GEO, and CPTAC-GDC cohorts; achieved 99.59% accuracy on a blind microarray cohort and 98.24% on external RNA-seq data.
- Built a 680-gene cross-platform panel and selective prediction workflow reaching 99.83% accuracy at 85.88% coverage on the evaluated external panel.
- **A17:** Designed and trained a residual MLP using 24 engineered features from seven epigenomic tracks and genomic-distance terms to predict Hi-C contact structure under strict chromosome-level holdouts and a locked test set.
- Found that relative chromatin structure generalized across unseen chromosomes more consistently than absolute contact-map scale; documented failed target-blind calibration and negative results in a manuscript in preparation.

Data Analyst Intern, Health, Safety & Environment

Jul 2026 – Aug 2026

Engro Technical Services

Karachi, Pakistan

- Developed Python, FastAPI, and Power Query pipelines for risk scoring and scheduled analytics across 130,000+ operational records spanning incidents, near misses, audits, inspections, waivers, and corrective actions.
- Automated daily change monitoring across approximately 1,750 controlled engineering and HSE files; built executive Power BI semantic models and DAX measures for operational decision support.

Undergraduate Researcher, Ichthyology Lab

Spring 2026 – Present

University of Southern Mississippi

Hattiesburg, MS

- Build reproducible Python workflows and chromosome-correspondence matrices to analyze chromosome-scale *Fundulus* genome assemblies, synteny, and candidate fusion/fission patterns.

Software Engineering Intern

Aug 2023 – Feb 2024

Shahid I.T. Consultants

Karachi, Pakistan

- Automated recurring record-update workflows in Python, reducing manual processing by 30%; supported integration testing, deployment documentation, and client application maintenance.

SELECTED ENGINEERING & RESEARCH SOFTWARE

BioLitGraph — Biomedical Literature Graph Explorer | Python, Flask, PubMed/PubTator, NetworkX | [GitHub](#)

2026

- Built a literature-mining system that retrieves biomedical papers, normalizes extracted entities into evidence-linked relationship graphs, and exports structured analysis tables as CSV/JSON.

AEGIS — Controlled Document Intelligence & Governance | Python, FastAPI, React, TypeScript, Tauri, SQLite | [GitHub](#)

2026

- Engineered a local-first Windows system with deterministic multi-format diffing, trusted-state reconciliation, SHA-256 version integrity, governance workflows, 55 regression/adversarial tests, and native installer packaging.

HEARTLINE — Native Android Lyric Companion | Kotlin, Jetpack Compose, Room, Android Services | [GitHub](#)

2026

- Engineered a native Android system with media-session detection, synchronized playback state, offline persistence, resilient service recovery, on-device image rendering, automated tests, Android lint, and CI builds.

QTL Analysis Toolkit | Python, pandas, Matplotlib | [GitHub](#)

2025

- Implemented LOD scans, permutation-based significance thresholds, significant-marker exports, and publication-ready statistical-genomics plots.

TECHNICAL SKILLS

Programming: Python, C++, SQL, R; Kotlin, Dart

Machine Learning & Scientific Computing: PyTorch, scikit-learn, pandas, NumPy, SciPy; classification, feature engineering, model evaluation

Research & Data: RNA-seq, microarrays, Hi-C, ATAC-seq, epigenomics; TCGA, GEO, CPTAC, ENCODE

Engineering: Linux/WSL, Git/GitHub, GitHub Actions, Docker, FastAPI, Flask, React, TypeScript, SQLite, SQLAlchemy, Power BI

LEADERSHIP & HONORS

President & Founder, Student Poets Association and Creative Writing Club, USM — Lead arts programming, public showcases, open mics, and club operations.

Selected Honors: Outstanding Cambridge Learner Award; ITC 371 Best Project & Best Assistant Project Manager; UGS Best Art as Research Project (2026); Best Arts Showcase for Poetry (2025, 2026); President's List.